

Stanley College of Engineering and Technology for Women(A)

B.E I Semester (Backlog) Examinations May-June-2023

Mathematics-I
(Common to all branches)

Time: 3 hours

Max.Marks:60

PART-A

Note: Answer all questions (Compulsory)

5 X 2=10M

- | | | | |
|---|---|-----|-----|
| 1 | Write the statement of Cauchy's nth root test | 2 M | L 1 |
| 2 | Verify Rolle's theorem for $f(x) = x^3$ in $[1,3]$ | 2 M | L 1 |
| 3 | Find first and second order partial derivatives of $x^3 + y^3 - 3axy$ | 2 M | L 1 |
| 4 | Find the Curl f if $f = xyi + yzj + zxk$ | 2 M | L 1 |
| 5 | Find the first two iterations for $x^3 - 2x - 5 = 0$ by the method of Bisection correct to three decimal places | 2 M | L 2 |

PART-B

Note: Answer all questions

5 X 10= 50M

- | | | | |
|---|---|------|----|
| 6 | a. Test the convergence of the series $\sqrt[n]{n+1} - \sqrt[n]{n-1}$ | 5 M | L3 |
| | b. Test the convergence of the series $\sum (nx/(n+1))^n$ | 5 M | L3 |
| | OR | | |
| | c. Test the convergence of the series $\sum (1.2.3.....n)/(3.5.7.....(2n+1))$ | 5 M | L3 |
| | d. Examine the convergence of the series $1 - 1/\sqrt{2} + 1/\sqrt{3} - 1/\sqrt{4} + \dots$ | 5 M | L3 |
| 7 | a. Prove that $\pi/6 + 1/(5\sqrt{3}) < (\sin)^{-1}(3/5) < \pi/6 + 1/8$ | 5 M | L4 |
| | b. Verify Cauchy's Mean value theorem for $f(x) = \sin x$ and $g(x) = \cos x$ on $(0, \pi/4)$ | 5 M | L3 |
| | OR | | |
| | c. Show that the radius of curvature at any point of the cycloid $x = (a + a\cos\theta)$ and $y = a(1 - \cos\theta)$ is $4a \cos(\theta/2)$ | 5 M | L4 |
| | d. Find the evloep of a system of concentric and coaxial ellipses of constant area. | 5 M | L4 |
| 8 | a. Find the maximum and minimum distances of the point $(3,4,12)$ from the sphere $x^2 + y^2 + z^2 = 4$ | 10 M | L5 |

OR

1

P.T.O

- c. If $u = F(x-y, y-z, z-x)$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ 5M L3
- d. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ then show that $(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z})^2 u = (-9)/(x+y+z)^2$ 5M L4
- 9 a. If $F = (x+y+1)i + j - (x+y)k$, show that $\text{Curl } F = 0$ 5M L3
- b. Show that the vector $F = 3y^4z^2i + 4x^3z^2j + 3x^2y^2k$ is solenoidal. 5M L3
- OR
- c. Verify Green's theorem for $\int_C (xy + y^2)dx + x^2 dy$ where C is bounded by $y = x$ and $y = x^2$ 10M L4
- 10 a. Apply Gauss elimination method to solve the equations $x + 4y - z = -5$, $x + y - 6z = -12$, $3x - y - z = 4$ 5M L3
- b. Find the root of the equation $\cos x = x e^x$ using the regula-falsi method correct to three decimal places. 5M L3
- OR
- c. Find the positive root of $x^4 - x = 10$ correct to three decimal places, using Newton-Raphson method. 5M L3
- d. Find the interpolating polynomial for (0,2), (1,3), (2,12) and (5,147) using Lagrange's interpolating formula. 5M L2



Hall Ticket No-

Code: S123513

Stanley College of Engineering and Technology for Women(A)

B.E.(CSE,CME,AI&DS) I Semester (Backlog) Examinations May-June-2023

Applied Physics

Time: 3 hours Max.Marks:60

PART-A

Note: Answer all questions (compulsory) 5 X 2=10M

- | | | | |
|---|--|-----|----|
| 1 | List out any four advantages of optical fibres over conventional communication channels. | 2 M | L1 |
| 2 | Define anti-ferro and ferri-magnetic materials | 2 M | L2 |
| 3 | Mention any four applications of superconductors | 2 M | L1 |
| 4 | Write a note on thermistor | 2 M | L1 |
| 5 | Classify different types of carbon nanotubes | 2 M | L1 |

PART-B

Note: Answer all questions 5 X 10= 50M

- | | | | |
|-----------|--|-----|----|
| 6 | a. Sketch a neat energy level diagram and explain principle, working and construction of a Ruby laser. | 7 M | L3 |
| | b. Differentiate between stimulated and spontaneous emissions. | 3 M | L2 |
| OR | | | |
| | c. Segregate optical fibres into different types and draw optical paths followed by light rays in them. | 6 M | L6 |
| | d. The refractive index core of an optical fibre is 1.625 and critical angle is 56.1° . Find the refractive index of Cladding, numerical aperture, and acceptance angle of the optical fibre. | 4 M | L2 |
| 7 | a. Define electronic polarization. Show that the electronic polarizability is proportional to cube of atomic radius. | 7 M | L6 |
| | b. Discuss ferro electricity in BaTiO_3 | 3 M | L2 |
| OR | | | |
| | c. Enlist the characteristics of ferro magnetism? Arrive at the expression for Curie temperature of a ferromagnetic material using Weiss theory. | 6 M | L5 |
| | d. Give any two differences between soft and hard magnetic materials. | 4 M | L3 |
| 8 | a. Summarise the general properties of Superconductors. | 6 M | L4 |

- b. The critical magnetic field of Vanadium at OK is 0.1 Tesla and critical temperature is 5K. Calculate critical magnetic field at 3K. 4 M L3

OR

- c. Deduce Eigen functions and Eigen values of a particle of mass 'm' with potential energy 'V' confined to a box of width 'L'. 6 M L4
- d. Estimate the wavelength associated with a ball of mass 20gm moving with a velocity of 200m/s and an electron accelerated under a potential difference of 200V. Compare the results. 4 M L3

- 9 a. Compute mathematically existence of energy bands in solids using Kronig penny model and give its findings. 7M L3
- b. Draw the energy band diagrams of conductors, semiconductors and insulators 3M L1

OR

- c. Derive expression for carrier concentration of an intrinsic semiconductor at a temperature T. 6 M L3
- d. State Hall effect and its mention applications. 4 M L1

- 10 a. Compare merits and demerits of thermal and electron beam evaporation techniques. With a good illustration discuss thermal evaporation method to prepare thin films. 7 M L4
- b. Write a note on working of solar cell 3 M L2

OR

- c. Explain Chemical vapour deposition technique to synthesis nano materials 6 M L4
- d. Mention the advantages and disadvantages of nanomaterials. 4 M L2
